QUESTION 1

1. Build a comparison table for the five models

Comparison of the five software development models

Model	Flexibility	Risk handling	Client involvement	Documentation	Cost of change	Advantages	Disadvantages	Example
Waterfall	Low	Weak	Low	Very high	High	Clear stages, predictable milestones, and detailed documentation.	Changes are difficult after a phase is complete; users see working software late.	A payroll system with stable, well-defined requirements.
Incremental	Medium	Medium	Medium	Medium	Medium	Delivers useful parts early; each increment can be tested and improved.	Needs careful architecture and planning; integrating increments can be challenging.	A course registration system released in stages: catalog, registration, then reports.
Prototype	High	Medium	Very high	Low	Low early on	An early model helps users explore screens and clarify unclear requirements.	Users may mistake the prototype for a finished product; rushed design may not scale.	A research app mock-up reviewed by researchers before full development.
Spiral	High	Strong	High	High	Medium	Risk analysis and stakeholder review happen throughout repeated cycles.	Can be costly and complex; requires risk-management expertise.	A banking platform where security and transaction risks need repeated review.
Agile	Very high	Good	Very high	Light	Low	Frequent working releases, continuous feedback, and easy adaptation to change.	Needs regular stakeholder participation and close team collaboration.	A small website delivered in short iterations and refined with client feedback.

QUESTION 2

Choose and justify a model for 4 scenarios

Banking system, small website, research prototype, and safety-critical system

Scenario	Recommended model	Justification
Banking system	Spiral	Banking software handles sensitive data and financial transactions. Spiral repeatedly identifies and reduces security, privacy, and transaction risks while allowing stakeholder review during development.
Small website	Agile	A small website can be built in short cycles. The team can release core pages early and adjust layout or features quickly using client feedback.
Research prototype	Prototype	Research needs may be unclear at first. A working prototype lets researchers try the idea, give feedback, and refine requirements before the full system is built.
Safety-critical system	Waterfall	A safety-critical project benefits from stable requirements, formal reviews, detailed documentation, and traceability through planned stages. Requirements changes must be controlled carefully.

QUESTION 3

Choose and justify the model for your team project

Recommended model: Agile

Agile is suitable for UniPlan because student and administrator needs can be refined as the team reviews working features. The team can develop in increments: student accounts and profiles, study-plan progress, course search, registration checks, and administrator tools. Feedback after each increment helps improve usability and clarify registration rules. This approach also lets the team test important rules, such as prerequisites, credit-hour limits, schedule conflicts, and section capacity, as those features are added.